Ethernet standard and Specification

1. RJ-45 Pin Layout and Color Codes (T568A vs. T568B)

When looking at the bottom of an RJ-45 modular plug with the **gold contact pins facing up towards you** and the plastic retaining clip facing down, pins are numbered **1 through 8 from left to right**.



Pinout and Color Assignment Table

Pin #	T568A Color Standard	T568B Color Standard	Fast Ethernet Signal (10/100 Mbps)	Gigabit Role (1000BASE-T)
Pin 1	**White / Green Stripe**	**White / Orange Stripe**	Transmit Positive (Tx+)	Bi-directional Pair A+
Pin 2	**Solid Green**	**Solid Orange**	Transmit Negative (Tx-)	Bi-directional Pair A-
Pin 3	**White / Orange Stripe**	**White / Green Stripe**	Receive Positive (Rx+)	Bi-directional Pair B+
Pin 4	**Solid Blue**	**Solid Blue**	Not used (or PoE DC $+$)	Bi-directional Pair C+
Pin 5	**White / Blue Stripe**	**White / Blue Stripe**	Not used (or PoE DC $+$)	Bi-directional Pair C-
Pin 6	**Solid Orange**	**Solid Green**	Receive Negative (Rx-)	Bi-directional Pair B-
Pin 7	**White / Brown Stripe**	**White / Brown Stripe**	Not used (or PoE DC $-$)	Bi-directional Pair D+
Pin 8	**Solid Brown**	**Solid Brown**	Not used (or PoE DC $-$)	Bi-directional Pair D-

Key Difference: T568A and T568B differ only in swapping the **Green pair (Pins 1 & 2)** with the **Orange pair (Pins 3 & 6)**. The Blue (Pins 4 & 5) and Brown (Pins 7 & 8) pairs remain identical.

2. Comparison Table: Ethernet Standards Specifications

Specification	10BASE-T	100BASE-TX (Fast Ethernet)	1000BASE-T (Gigabit Ethernet)
IEEE Standard	IEEE 802.3i	IEEE 802.3u	IEEE 802.3ab
Data Speed (Throughput)	**10 Mbps**	**100 Mbps**	**1,000 Mbps (1 Gbps)**
Baseband Signaling	Manchester Encoding	4B5B / MLT-3 Encoding	PAM-5 (5-level Pulse Amplitude)
Minimum Cable Type	Cat3 UTP (or Cat5)	Cat5 UTP (or Cat5e)	Cat5e UTP (or Cat6)
Wire Pairs Used	2 Pairs (4 wires: Pins 1, 2, 3, 6)	2 Pairs (4 wires: Pins 1, 2, 3, 6)	All 4 Pairs (8 wires: full-duplex)
Maximum Cable Length	**100 meters** (328 feet)	**100 meters** (328 feet)	**100 meters** (328 feet)
Connector Type	8P8C (RJ-45)	8P8C (RJ-45)	8P8C (RJ-45)

3. Scenario: Connecting Two Laptops Directly via Ethernet

Which Cable to Use:

- **Historically (Theoretical / CompTIA standard):** You must use an **Ethernet Crossover Cable** (one end wired with **T568A** and the other with **T568B**).
- **Modern Reality:** A standard **Straight-Through Cable** (both ends T568B) will work seamlessly on any modern laptop due to **Auto-MDI/MDIX**.

Why (Technical Reasoning):

1. **The Transmit / Receive Problem:** Network Interface Cards (NICs) on client devices (MDI devices) transmit data on **Pins 1 & 2** and receive incoming data on **Pins 3 & 6**.
2. **If using a Straight-Through Cable without Auto-MDIX:** Laptop A's transmit pins (Pins 1 & 2) connect directly to Laptop B's transmit pins (Pins 1 & 2), causing a transmission collision. Neither laptop receives any data because both listening receive pins (Pins 3 & 6) remain idle.
3. **How a Crossover Cable Solves It:** By terminating one end in T568A and the other in T568B, the Orange and Green pairs cross over:
    - Laptop A Tx (Pins 1 & 2) $\rightarrow$ connects physically to Laptop B Rx (Pins 3 & 6).
    - Laptop B Tx (Pins 1 & 2) $\rightarrow$ connects physically to Laptop A Rx (Pins 3 & 6).
4. **Auto-MDI/MDIX Detection:** Modern Gigabit (1000BASE-T) Network Interface Cards incorporate automatic medium-dependent interface crossover (Auto-MDIX). The chip detects the pinout electronically and swaps transmit and receive signals internally, allowing a normal straight-through cable to function directly between two PCs.

4. Three Real-World Devices & Supported Ethernet Standards

Device	Location	Likely Supported Standard	Explanation / Reason
1. Desktop PC / Gaming Laptop	Home / Lab Desk	**1000BASE-T (Gigabit Ethernet)**	High-bandwidth motherboard NICs require Gigabit (1 Gbps) speeds to transfer multi-gigabyte files across local shares, play online multiplayer games with low latency, and utilize fast broadband connections.
2. Network Laser Printer / Photocopier	College Computer Lab	**100BASE-TX (Fast Ethernet)**	Printers only receive small spool files, document raster images, and status packets (a few megabytes at most). A 100 Mbps connection easily provides sufficient bandwidth at a lower manufacturing component cost.
3. Home Wi-Fi 6 Router (WAN / LAN Ports)	Living Room / Office Desk	**1000BASE-T (Gigabit)** *(or 2.5GBASE-T)*	The router must route multi-hundred megabit fiber internet traffic to several wireless and wired devices simultaneously without becoming a physical network bottleneck.